

Study	Architecture	Control / sensing	Cooling / recovery	Relevance to this thesis
Copaci et al.	Opposing flexible SMA/Bowden actuators; upper-limb exoskeleton	Bilinear PID + PWM + joint-position feedback	Natural convection	Direct actuator/control lineage; different joint and loading.
Hau et al.	Opposed SMA linear actuators acting on an ankle footplate; stationary platform	Discrete state/threshold control + position/force feedback	Forced air	Direct antagonistic ankle precedent; non-wearable and actively cooled.
Mansilla Navarro et al.	Wearable soft/exosuit; opposing Bowden/PTFE SMA actuators	Two-level control + IMU feedback + bilinear low-level regulation	Passive cooling with recovery intervals	Closest wearable control lineage; trajectory period strongly affects error.
Zhu et al.	Opposing SMA wire groups with differential pulleys	PI closed-loop angular control	Forced air	Useful angular-error benchmark under different geometry, load, and cooling.
Present thesis	Adjustable wearable ankle exoskeleton; antagonistic three-wire groups + Bowden/PTFE	Relative-angle feedback + State Manager + BPID + Safety Supervisor + final gating	Passive recovery opportunities + manually scheduled long-duration recovery	Mannequin-based evaluation: transmission sensitivity, repeatability, and 38-min active/recovery operation.

Contextual positioning only – not a performance ranking. Geometry, load, cooling, sensing, trajectory, and validation protocols differ across studies.